Telecommunications Application Project

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Introduction

In this project the purpose was to take data from a 3-axis-accelometer and transport and store the data using different technologies (see Figure 1 for a clear picture). Finally, algorithms were used to analyze the data.

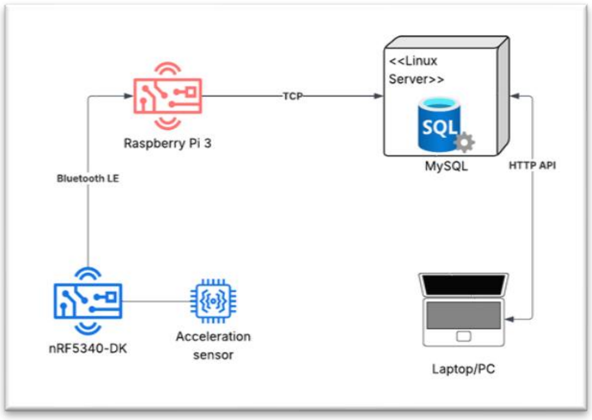

FIGURE 1. Project Architecture

Objective

The objective for this project was to learn how to utilize previously learned topics with a real-world application. These previously learned topics include Bluetooth Low Energy (BLE), Machine Learning (ML) and TCP protocols. Additionally, this project served as an introduction to Linux servers.

Methods

To capture XYZ–directional data from the sensor, a Nordic nRF5340 Development Kit was used (Figure 2).

With code written in C, the captured data was sent via BLE to a Raspberry Pi 3.On the Raspberry, a program written with Python, packaged the data and sent it to a Linux server via TCP. The Ubuntu Linux server was configured with a MySQL database. From the database, the data could be fetched with HTTP queries and analyzed. A k-means algorithm was written using python. This algorithm found approximate center points from samples that were taken from 6 directions, with the sensor pointing up, down, left, right, forwards and backwards (Figure 3).

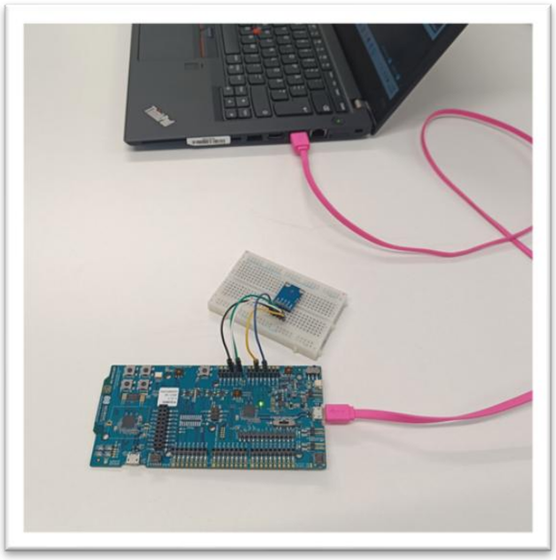

FIGURE 2. The 3-axis-accelometer and the nRF5340

Results

The project worked as expected. We could capture a hundred sensor readings with a single button press, which were automatically sent to the database.

To check the center points validity, they were checked by making a confusion matrix on the nRF5340 and checked with actual data readings. The program could tell which way our sensor was being held.

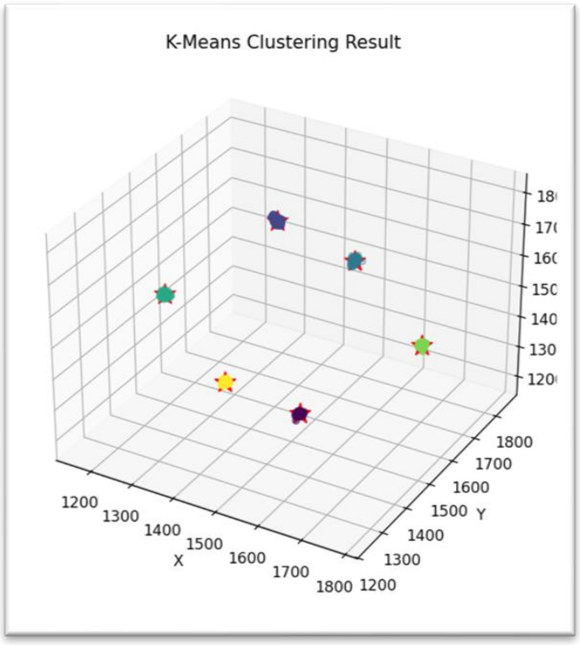

FIGURE 3. K-Means algorithm result. The red stars are the calculated center points for the data clusters (colored). Each cluster is made of 100 data points.

Conclusions

This project could have been a little more cohesive to be more like a working product, but the assignment was given to us. Yet, the necessary tasks were completed, and the end result was satisfactory. A lot was learned during the project, especially about Python and how to handle data.